

# RealView® Tools for System Design Optimization

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## System Level Design

- Simultaneous hardware and software co-design critical for optimal system design
- ARM is seeing that
  - Software designed in parallel with hardware improves power consumption and execution performance
  - Building software and hardware in silos leads to inefficient designs
  - Complex systems with multiple processors in a system need careful software design
    - Multi-threading
    - Power-saving strategies
  - Tools that foster interaction of the hardware and software design teams yields a better time to market and more cost-efficient implementation
    - System Design Optimization (SDO) is the key to next-generation mobile, consumer, and embedded devices



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## Trends

- Industry is looking for better hardware/software tools solutions
- Traditional sequential model of developing hardware first, software second are no longer yielding competitive results
- Early attempts at useful system design observed with extending ARMulator with handwritten peripherals in C
  - Major technical drawbacks
    - Resulting efficiency only provided reasonable results after lots of tweaking and careful statistical analysis (very time consuming!)
    - But, reinforced a very potent message: system design is key
- Industry is looking to fast, accurate system models as a way to bridge the divide between hardware and software design team silos
- Better solutions than ARMulator are necessary to enable complex designs to evolve rapidly and reliably



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## Typical Design Problems

- Difficult to map use cases to actual hardware design
- Mistakes in estimation of power or MIPS budget are very costly:
  - Over-engineering or under-engineering designs can cost millions
- Software development usually late
  - Often due to inadequate early representations of final system design
  - Defects are more to repair in hardware late in the design stage
- A more evolved, holistic approach to system design is needed:
  - RealView tools represent the most mature, connected solution for System Design Optimization (SDO)
  - Focus on providing an integrated solution: hardware design and software development suite, enabling pre-silicon development through applications development on real hardware



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## Paradigm Shift in System Level Design



- Virtual Phase
  - Architecture exploration
  - Trade-off analysis
  - IP selection
  - Pre-silicon SW development
  - HW and SW optimization
- Real-Life Phase
  - HW implementation
  - SW development
  - HW/SW co-verification

→ ESL based

*The Virtual Phase increasingly determines overall product success*



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## Samsung Proof Point

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**Samsung's VIP Design Methodology Reduces SoC Design Time Up to 40 Percent**

SEOUL, Korea - September 14, 2004 - Samsung Electronics Co., Ltd., a leader in advanced semiconductor technology, today revealed a hardware/software co-design methodology that will reduce overall design time for the company's System-on-Chip (SoC) products by up to 40 percent. The Virtual IP (VIP) approach will ultimately result in faster production of Samsung logic components while providing customers a time-to-market advantage.

Samsung developed the VIP design methodology in partnership with EDA vendors to address the ever-increasing complexity of SoC design challenges. Today's exploding market for multi-function consumer products requires more on-chip firmware as well as transistors, complicating and extending the design process. Using the VIP methodology, simulation speeds are hundreds or even thousands of times faster than that of a traditional RTL approach while maintaining more than 90 percent of accuracy over RTL.

With the conventional RTL design methodology, it was not possible to design and optimize software before the field programmable gate array (FPGA) prototype is made. However, the VIP methodology enables the error-free design of software before the development of RTL code, reducing the SoC design time by up to 40 percent.



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## System Design Optimization Lets You...

- Perform trade-off analysis of different system components
  - Cores, bus architectures, caches, memories
- Develop hardware and software architectures together
  - Profile hardware and software performance and iterate on design
    - Develop optimal software architecture and algorithms for hardware
    - Develop optimal hardware architecture to run the software
- Deliver cycle accurate and cycle approximate system model
  - Develop firmware, device drivers, port OS and profile results
- Deliver very high performance Real-Time System Models (RTSM)
  - Develop application before silicon with real-time performance
- Commit to RTL considerably later in the design
  - More optimal design due to iteration
  - Higher confidence in performance expectations
  - Less time spent rewriting software to fit flaws in the design
  - Up to 40 percent time-to-market advantage!



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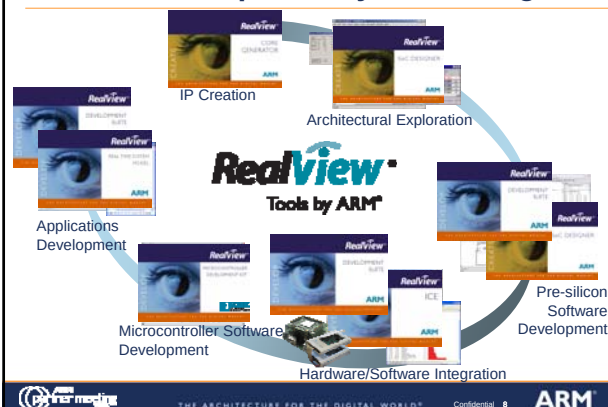
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## RealView for Optimal System Design



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## Unique Value Proposition

RealView tools by ARM are unique in their ability to provide solutions that span the complete development process from concept to final product deployment.

- RealView DEVELOP and CREATE families of tools offer
  - Fast route to IP evaluation and system prototyping
  - Total solution for hardware/software co-development
  - Integrated solution of interoperable tools that collaborate and cooperate
  - This unique ability leads to:
    - Enhanced interdisciplinary communication
    - The ability to find design flaws at any point in the design flow



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## When Do You Find Flaws?

- Not all design flaws introduced in a development phase can be found in *that* phase
  - Design flaws in software can now get caught before silicon
- But...
  - Design flaws in hardware may only manifest themselves
    - When software architecture is being tested
    - When the firmware is being created
    - Or worst of all...When the software application is running!
- Only the RealView Tools provide developers with the ability to go forward and backward through the design cycle to find flaws



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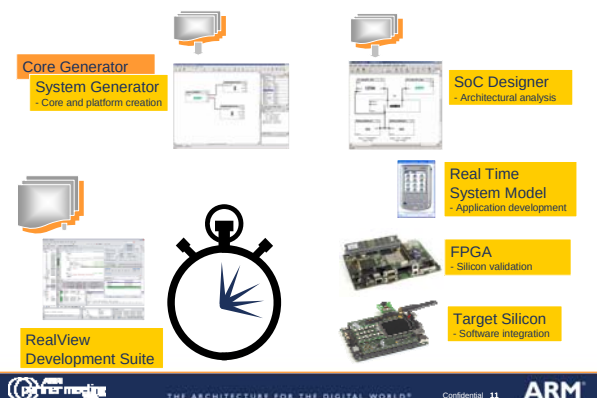
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## Moving Forward Through the Cycle



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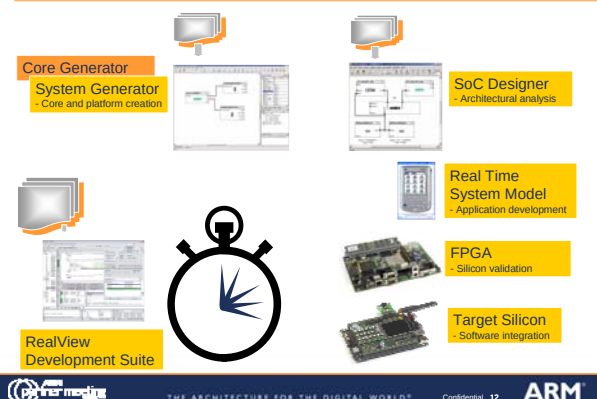
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## Going Backwards Through the Cycle



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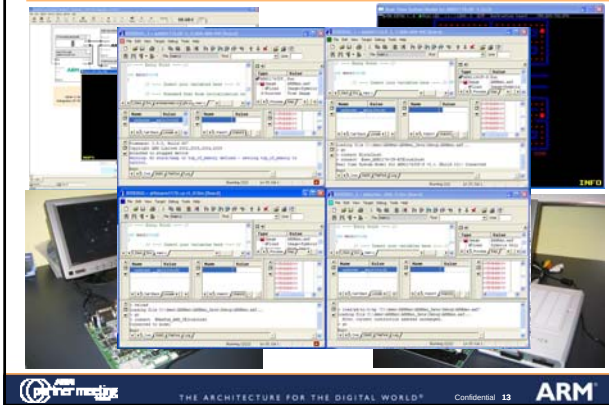
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## Four Targets With a Single Debugger...




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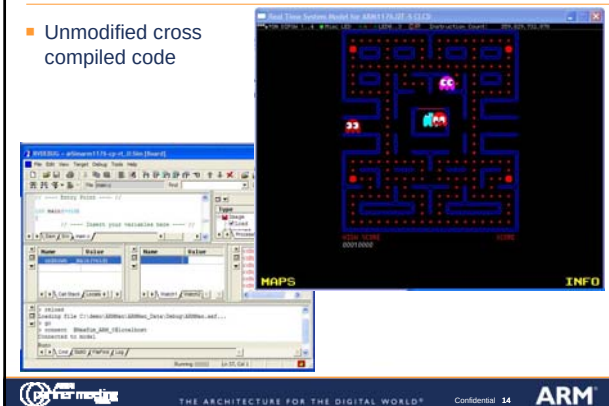
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## ...and a Single Target Binary

- Unmodified cross compiled code




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